1.9.2 Line insulator materials

Porcelain, glass, magnesium silicate etc. are used to produce line insulators. The porcelain is used to produce the insulating material. It is made by suitable heat in combination with plastic, white clay and glass. It is also made with good stability and smooth surface and free from porosity.

1.9.3 Types of insulators

i. Pin type insulator
ii. Suspension insulators
    a. Hewlett suspension type
    b. Cemented cap type
    c. Core and link type
iii. Strain insulator
iv. Shackle insulator
v. Stay type insulator

i. Pin type insulator



Fig. 1.10 **Pin type insulator**

This type of insulator is fixed in the crossing arm of the pole as shown in figure 1.10. The conductor is placed in the top of semicircular groove and the conductor is placed on it. This type of insulator is used to capture straight conductors

ii. Suspension insulators

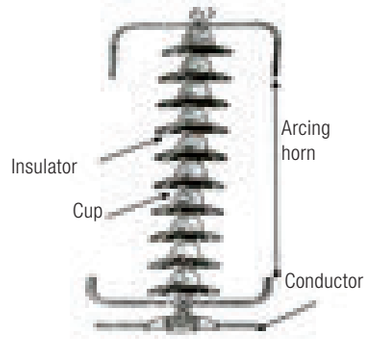


Fig. 1.11 **Suspension type insulator**

The suspension type line insulator is shown in figure 1.11. This insulator is in hanging shape and is connected to the steel tower. The line conductor is connected in the base. In suspension type insulator, a number of similar units are connected one by one with bi-metallic links. Each suspension insulator is designed for 11KV. Therefore, by connecting a number of such insulator discs, a string of insulator can be designed for any required voltage.

a. Hewlett type



Fig. 1.12 **Hewlett type insulator**

The Hewlett type line insulator is shown in figure 1.12. The insulator part

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